FINANCING CHOICES

- You are reading the Wall Street Journal and notice a tombstone ad for a company, offering to sell convertible preferred stock. What would you hypothesize about the health of the company issuing these securities?
  a. Nothing
  b. Healthier than the average firm
  c. In much more financial trouble than the average firm

Bed Bath & Beyond Inc. Announces Proposed Offering of Series A Convertible Preferred Stock and Warrants

CAPITAL STRUCTURE: FINDING THE RIGHT FINANCING MIX

You can have too much debt… or too little..

THE BIG PICTURE..

Maximize the value of the business (firm)

The Investment Decision
Invest in assets that earn a return greater than the minimum acceptable hurdle rate

- The **hurdle rate** should reflect the riskiness of the investment and the mix of debt and equity used to fund it.
- The **return** should reflect the magnitude and the timing of the cashflows as well as all side effects.

The Financing Decision
Find the right kind of debt for your firm and the right mix of debt and equity to fund your operations

- The **optimal mix** of debt and equity maximizes firm value
- The **right kind** of debt matches the tenor of your assets

The Dividend Decision
If you cannot find investments that make your minimum acceptable rate, return the cash to owners of your business

- **How much cash** you can return depends upon current & potential investment opportunities
- **How you choose** to return cash to the owners will depend on whether they prefer dividends or buybacks

PATHWAYS TO THE OPTIMAL

1. **The Cost of Capital Approach**: The optimal debt ratio is the one that minimizes the cost of capital for a firm.
2. **The Enhanced Cost of Capital approach**: The optimal debt ratio is the one that generates the best combination of (low) cost of capital and (high) operating income.
3. **The Adjusted Present Value Approach**: The optimal debt ratio is the one that maximizes the overall value of the firm.
4. **The Peer Group Approach**: The optimal debt ratio is the one that brings the firm closes to its peer group in terms of financing mix.
5. **The Life Cycle Approach**: The optimal debt ratio is the one that best suits where the firm is in its life cycle.

I. THE COST OF CAPITAL APPROACH

- Value of a Firm = Present Value of Cash Flows to the Firm, discounted back at the cost of capital.
  - If the **cash flows to the firm are held constant**, and the cost of capital is minimized, the value of the firm will be maximized.
  - If the end game for a business is maximizing its value, its **optimal** debt ratio will be the **one that minimizes the cost of capital**.
- Cost of Capital = Cost of Equity (E/(D+E)) + Pre-tax Cost of Debt (1- tax rate) (D/(D+E)
- Within this framework, here are the key questions:
  a. As the debt ratio changes, how does the cost of capital change?
  b. And if the cost of capital does change, what is causing it to change?

THE DEBT TRADE OFF ON THE COST OF CAPITAL

As you change the debt ratio

Cost of Equity x Weight of equity + (Risk free Rate + Default Spread) x (1- Tax Rate) x Weight of Debt

Here is what happens to the costs of debt & equity

Cost of equity = Riskfree Rate + Levered Beta (Equity Risk Premium)

As you borrow more, your D/E ratio increases and as it does, so does your levered beta.

As interest coverage ratio increases, synthetic rating drops & default spread rises

As you borrow more, your interest expense rises and your interest coverage ratio decreases

Tax rate decreases if interest expense > operating income

When interest expenses rise above operating income, the tax benefit dissipates.

The trade off: As you use more debt, you replace more expensive equity with cheaper debt but you also increase the costs of equity and debt. The net effect will determine whether the cost of capital will increase, decrease or be unchanged as debt ratio changes.

COSTS OF DEBT & EQUITY

- An article in an Asian business magazine argued that equity was cheaper than debt, because dividend yields are much lower than interest rates on debt. Do you agree with this statement?
  a. Yes
  b. No
- Can equity ever be cheaper than debt?
  a. Yes
  b. No

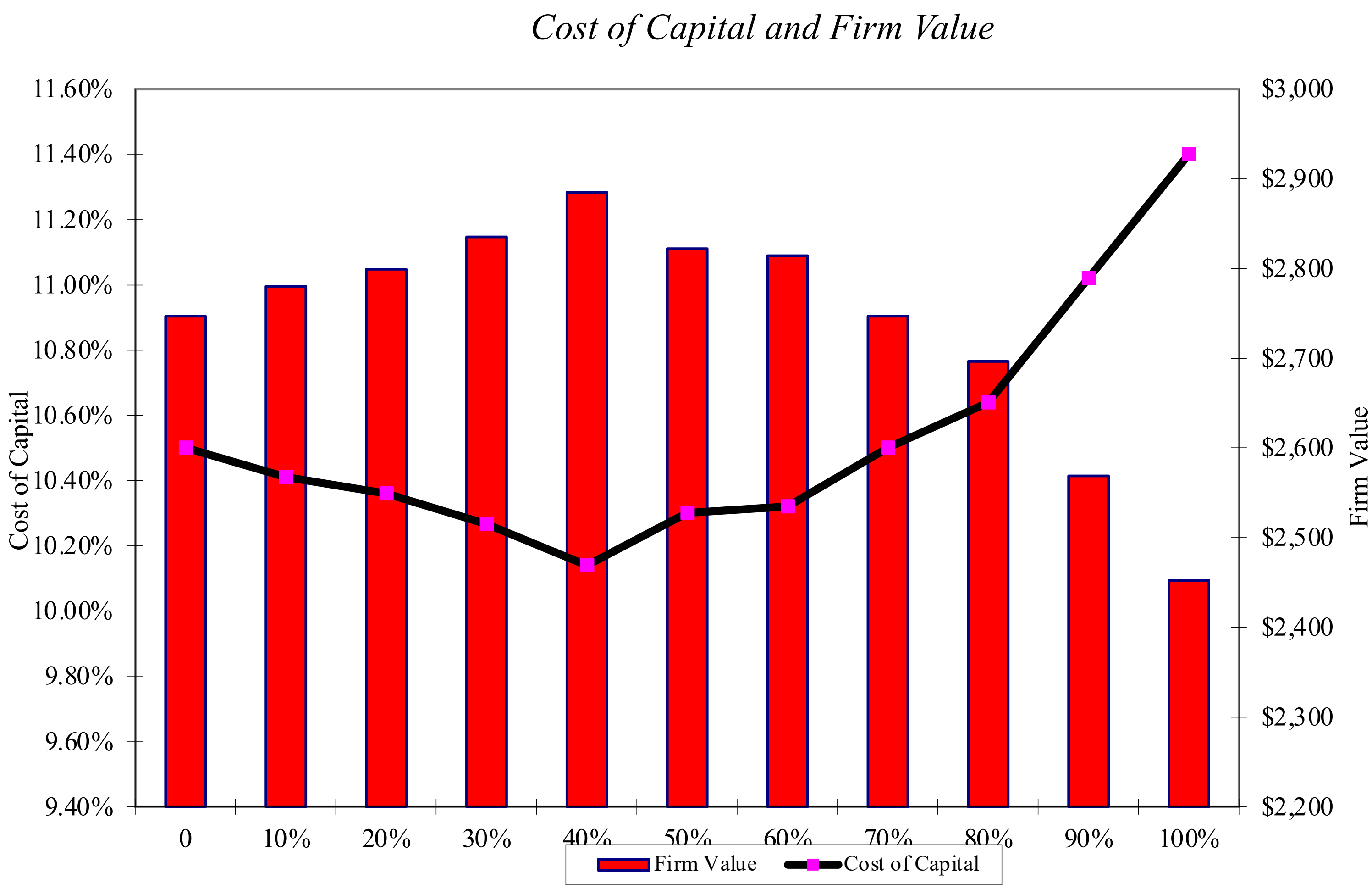
APPLYING COST OF CAPITAL APPROACH: THE TEXTBOOK EXAMPLE

Assume the firm has \$200 million in cash flows, expected to grow 3% a year forever.

D/(D+E)	Cost of Equity	After-tax Cost of Debt	Cost of Capital	Firm Value
0	10.50%	4.80%	10.50%	\$2,747
10%	11.00%	5.10%	10.41%	\$2,780
20%	11.60%	5.40%	10.36%	\$2,799
30%	12.30%	5.52%	10.27%	\$2,835
40%	13.10%	5.70%	10.14%	\$2,885
50%	14.50%	6.10%	10.30%	\$2,822
60%	15.00%	7.20%	10.32%	\$2,814
70%	16.10%	8.10%	10.50%	\$2,747
80%	17.20%	9.00%	10.64%	\$2,696
90%	18.40%	10.20%	11.02%	\$2,569
100%	19.70%	11.40%	11.40%	\$2,452

$$\text{Value} = \frac{\text{Expected Cash flow to firm next year}}{(\text{Cost of capital } - g)} = \frac{200(1.03)}{(\text{Cost of capital } - g)}$$

THE U-SHAPED COST OF CAPITAL GRAPH…



CURRENT COST OF CAPITAL: DISNEY

- The **beta for Disney's stock in November 2013** was 1.0013. The T. bond rate at that time was 2.75%. Using an estimated equity risk premium of 5.76%, we estimated the cost of equity for Disney to be 8.52%:

  Cost of Equity = 2.75% + 1.0013(5.76%) = 8.52%

- Disney's **bond rating in May 2009 was A**, and based on this rating, the estimated pretax cost of debt for Disney is 3.75%. Using a marginal tax rate of 36.1, the after-tax cost of debt for Disney is 2.40%.

  After-Tax Cost of Debt = 3.75% (1 – 0.361) = 2.40%

- The cost of capital was calculated using these costs and the weights based on **market values of equity (121,878) and debt (15.961):**

$$\text{Cost of capital} = 8.52\% \frac{121{,}878}{(15{,}961+121{,}878)} + 2.40\% \frac{15{,}961}{(15{,}961+121{,}878)} = 7.81\%$$

MECHANICS OF COST OF CAPITAL ESTIMATION

1. Estimate the Cost of Equity at different levels of debt:
   - Equity will become riskier -> Beta will increase -> Cost of Equity will increase.
   - Estimation will use levered beta calculation
2. Estimate the Cost of Debt at different levels of debt:
   - Default risk will go up and bond ratings will go down as debt goes up -> Cost of Debt will increase.
   - To estimating bond ratings, we will use the interest coverage ratio (EBIT/Interest expense)
3. Estimate the Cost of Capital at different levels of debt
4. Calculate the effect on Firm Value and Stock Price.

LAYING THE GROUNDWORK:
1. ESTIMATE THE UNLEVERED BETA FOR THE FIRM

- **The Regression Beta**: One approach is to use the regression beta (1.25) and then unlever using the average debt to equity ratio (19.44%) during the period of the regression:
  - Unlevered beta = = 1.25 / (1 + (1 - 0.361)(0.1944))= 1.1119
- **The Bottom-up Beta**: Alternatively, we can back to the source and estimate it from the betas of the businesses.

Business	*Revenues*	*EV/Sales*	*Value of Business*	*Proportion of Disney*	*Unlevered beta*	*Value*	*Proportion*
Media Networks	$20,356	3.27	$66,580	49.27%	1.03	$66,579.81	49.27%
Parks & Resorts	$14,087	3.24	$45,683	33.81%	0.70	$45,682.80	33.81%
Studio Entertainment	$5,979	3.05	$18,234	13.49%	1.10	$18,234.27	13.49%
Consumer Products	$3,555	0.83	$2,952	2.18%	0.68	$2,951.50	2.18%
Interactive	$1,064	1.58	$1,684	1.25%	1.22	$1,683.72	1.25%
Disney Operations	**$45,041**		**$135,132**	**100.00%**	**0.9239**	**$135,132.11**	**100.00%**

2. Get Disney's current financials…

	Most recent fiscal year (2012-13)	Prior year
Revenues	$45,041	$42,278
EBITDA	$10,642	$10,850
Depreciation & Amortization	$2,192	$1,987
EBIT	$9,450	$8,863
Interest Expenses	$349	$564
EBITDA (adjusted for leases)	$12,517	$11,168
Depreciation (adjusted for leases)	$ 2,485	$2,239
EBIT (adjusted for leases)	$10,032	$8,929
Interest Expenses (adjusted for leases)	$459	$630

STEP 1: COST OF EQUITY

Debt to Capital Ratio	D/E Ratio	Levered Beta	Cost of Equity
0%	0.00%	0.9239	8.07%
10%	11.11%	0.9895	8.45%
20%	25.00%	1.0715	8.92%
30%	42.86%	1.1770	9.53%
40%	66.67%	1.3175	10.34%
50%	100.00%	1.5143	11.48%
60%	150.00%	1.8095	13.18%
70%	233.33%	2.3016	16.01%
80%	400.00%	3.2856	21.68%
90%	900.00%	6.2376	38.69%

Levered Beta = 0.9239 (1 + (1- .361) (D/E))
Cost of equity = 2.75% + Levered beta * 5.76%

STEP 2: ESTIMATING COST OF DEBT

- Start with the market value of the firm = = 121,878 + $15,961 = $137,839 million

D/(D+E)	0.00%	10.00%	Debt to capital
D/E	0.00%	11.11%	D/E = 10/90 = .1111
$ Debt	$0	$13,784	10% of $137,839
EBITDA	$12,517	$12,517	Same as 0% debt
Depreciation	$ 2,485	$ 2,485	Same as 0% debt
EBIT	$10,032	$10,032	Same as 0% debt
Interest	$0	$434	Pre-tax cost of debt * $ Debt
Pre-tax Int. cov	∞	23.10	EBIT/ Interest Expenses
Likely Rating	AAA	AAA	From Ratings table
Pre-tax cost of debt	3.15%	3.15%	Riskless Rate + Spread

THE RATINGS TABLE

Interest coverage ratio is	*Rating is*	*Spread is*	*Interest rate*
> 8.50	Aaa/AAA	0.40%	3.15%
6.5 – 8.5	Aa2/AA	0.70%	3.45%
5.5 – 6.5	A1/A+	0.85%	3.60%
4.25 – 5.5	A2/A	1.00%	3.75%
3 – 4.25	A3/A-	1.30%	4.05%
2.5 -3	Baa2/BBB	2.00%	4.75%
2.25 –2.5	Ba1/BB+	3.00%	5.75%
2 – 2.25	Ba2/BB	4.00%	6.75%
1.75 -2	B1/B+	5.50%	8.25%
1.5 – 1.75	B2/B	6.50%	9.25%
1.25 -1.5	B3/B-	7.25%	10.00%
0.8 -1.25	Caa/CCC	8.75%	11.50%
0.65 – 0.8	Ca2/CC	9.50%	12.25%
0.2 – 0.65	C2/C	10.50%	13.25%
<0.2	D2/D	12.00%	14.75%

T.Bond rate =2.75%

A TEST: CAN YOU DO THE 30% LEVEL?

		Iteration 1 (Debt @AAA rate)	Iteration 2 (Debt @AA rate)
D/(D + E)	*20.00%*	*30.00%*	*30.00%*
D/E	25.00%	30/70=42.86%	
$ Debt	$27,568	$41,352	
EBITDA	$12,517	$12.517	
Depreciation	$2,485	$2,485	
EBIT	$10,032	$10.032	
Interest expense	$868	41352*.0315=1,302	41352*.0345=1427
Interest coverage ratio	11.55	10032/1302=7.7	10032/1427=7.03
Likely rating	AAA	AA	AA
Pretax cost of debt	3.15%	3.45%	3.45%

BOND RATINGS, COST OF DEBT AND DEBT RATIOS

Debt Ratio	$ Debt	Interest Expense	Interest Coverage Ratio	Bond Rating	Pre-tax cost of debt	Tax rate	After-tax cost of debt
0%	$0	$0	∞	Aaa/AAA	3.15%	36.10%	2.01%
10%	$13,784	$434	23.10	Aaa/AAA	3.15%	36.10%	2.01%
20%	$27,568	$868	11.55	Aaa/AAA	3.15%	36.10%	2.01%
30%	$41,352	$1,427	7.03	Aa2/AA	3.45%	36.10%	2.20%
40%	$55,136	$2,068	4.85	A2/A	3.75%	36.10%	2.40%
50%	$68,919	$6,892	1.46	B3/B-	10.00%	36.10%	6.39%
60%	$82,703	$9,511	1.05	Caa/CCC	11.50%	36.10%	7.35%
70%	$96,487	$11,096	0.90	Caa/CCC	11.50%	32.64%	7.75%
80%	$110,271	$13,508	0.74	Ca2/CC	12.25%	26.81%	8.97%
90%	$124,055	$16,437	0.61	C2/C	13.25%	22.03%	10.33%

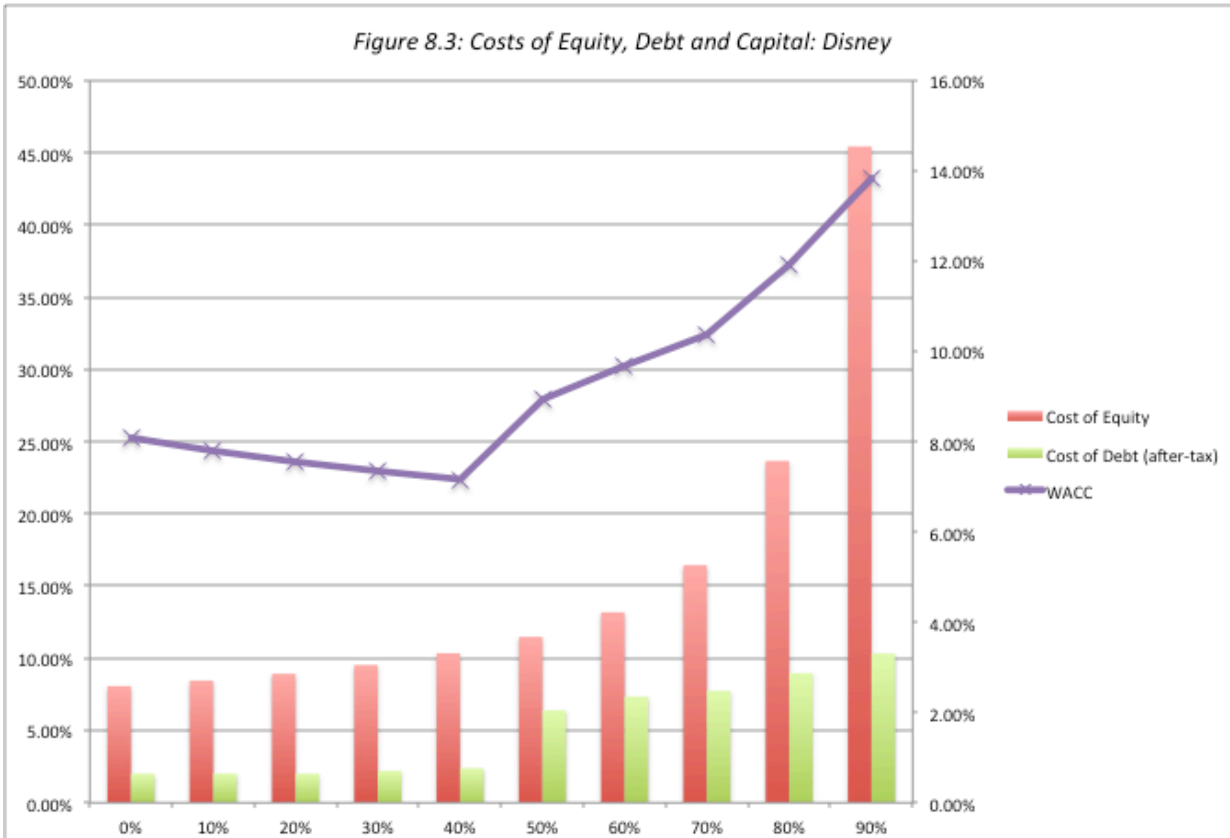
STATED VERSUS EFFECTIVE TAX RATES

- You need **taxable income for interest to provide a tax savings.** Note that the EBIT at Disney is $10,032 million. As long as interest expenses are less than $10,032 million, interest expenses remain fully tax-deductible and earn the 36.1% tax benefit. At an 60% debt ratio, the interest expenses are $9,511 million and the tax benefit is therefore 36.1% of this amount.
- At a **70% debt ratio, however, the interest expenses balloon to $11,096 million**, which is **greater than the EBIT of $10,032 million.** We consider the tax benefit on the interest expenses up to this amount:
  - Maximum Tax Benefit = EBIT * Marginal Tax Rate = $10,032 million * 0.361 = $ 3,622 million
  - Adjusted Marginal Tax Rate = Maximum Tax Benefit/Interest Expenses = $3,622/$11,096 = 32.64%

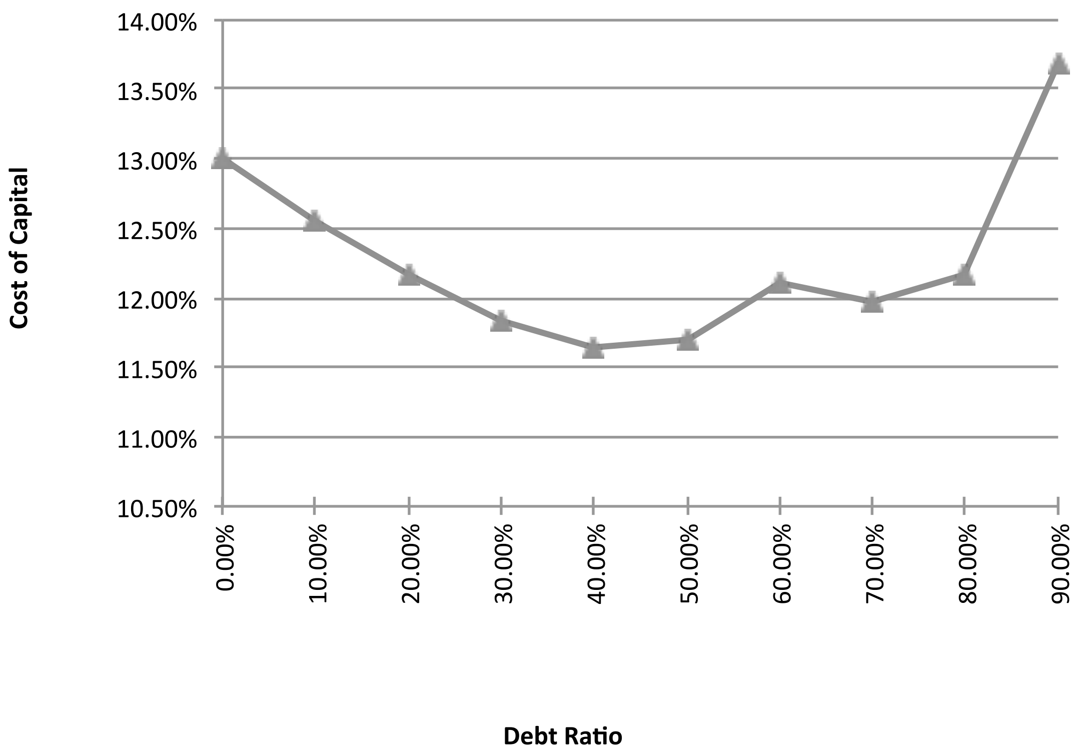
STEP 3: DISNEY'S COST OF CAPITAL SCHEDULE…

Debt Ratio	Beta	Cost of Equity	Cost of Debt (after-tax)	WACC
0%	0.9239	8.07%	2.01%	8.07%
10%	0.9895	8.45%	2.01%	7.81%
20%	1.0715	8.92%	2.01%	7.54%
30%	1.1770	9.53%	2.20%	7.33%
40%	1.3175	10.34%	2.40%	7.16%
50%	1.5143	11.48%	6.39%	8.93%
60%	1.8095	13.18%	7.35%	9.68%
70%	2.3762	16.44%	7.75%	10.35%
80%	3.6289	23.66%	8.97%	11.90%
90%	7.4074	45.43%	10.33%	13.84%

DISNEY: COST OF CAPITAL CHART



DISNEY: COST OF CAPITAL CHART: 1997



Note the kink in the cost of capital graph at 60% debt. What is causing it?

THE COST OF CAPITAL APPROACH SUGGESTS THAT DISNEY SHOULD DO THE FOLLOWING…

- Disney **currently has $15.96 billion in debt**. The optimal dollar debt (at 40%) is **roughly $55.1 billion**. Disney has excess debt capacity of 39.14 billion.
- To move to its optimal and gain the increase in value, Disney should borrow $ 39.14 billion and buy back stock.
- Given the magnitude of this decision, you should expect to answer three questions:
  a. Why should we do it?
  b. What if something goes wrong?
  c. What if we don't want (or cannot) buy back stock and want to make investments with the additional debt capacity?

WHY SHOULD WE DO IT? EFFECT ON FIRM VALUE – FULL VALUATION

- Step 1: Estimate the **cash flows to Disney as a firm**

EBIT (1 – Tax Rate) = 10,032 (1 – 0.361) =	$6,410
+ Depreciation and amortization =	$2,485
– Capital expenditures =	$5,239
– Change in noncash working capital	$0
Free cash flow to the firm =	$3,657

- Step 2: Back out **the implied growth rate** in the current market value
  - Current enterprise value = $121,878 + 15,961 - 3,931 = 133,908
  - Value of firm = $ 133,908 = $\frac{FCFF_0(1+g)}{(\text{Cost of Capital -g})} = \frac{3,657(1+g)}{(.0781 \text{ -g})}$
  - Growth rate = = (133,908* 0.0781 – 3,657)/(133,908+ 3,657) = 0.0494 or 4.94%
- Step 3: Revalue the firm with the new cost of capital
  - Firm value = $\frac{FCFF_0(1+g)}{(\text{Cost of Capital -g})} = \frac{3,657(1.0494)}{(.0716 \text{ -0.0484})} = \$172,935 \text{ million}$
  - Increase in firm value = $172,935 - $133,908 = $39,027 million

EFFECT ON VALUE: INCREMENTAL APPROACH

- In this approach, we start with the current market value and **isolate the effect of changing the capital structure** on the cash flow and the resulting value.
  - Enterprise Value before the change = $133,908 million
  - Cost of financing Disney at existing debt ratio = $ 133,908 * 0.0781 = $10,458 million
  - Cost of financing Disney at optimal debt ratio = $ 133,908 * 0.0716 = $ 9,592 million
  - **Annual savings** in cost of financing = $10,458 million – $9,592 million = $866 million

$$\text{Increase in Value} = \frac{\text{Annual Savings next year}}{(\text{Cost of Capital - g})} = \frac{\$866}{(0.0716 - 0.0275)} = \$19{,}623 \text{ million}$$

Enterprise value after recapitalization = Existing enterprise value + PV of Savings = $133,908 + $19,623 = $153,531 million

FROM FIRM VALUE TO VALUE PER SHARE: THE RATIONAL INVESTOR SOLUTION

- Because the increase in value accrues entirely to stockholders, we can estimate the increase in value per share by **dividing by the total number of shares outstanding (1,800 million).**
  - Increase in Value per Share = $19,623/1800 = $ 10.90
  - New Stock Price = $67.71 + $10.90= $78.61
- Implicit in this computation is the **assumption that the increase in firm value will be spread evenly across both the stockholders who sell their stock back to the firm and those who do not** and that is why we term this the **"rational"** solution, since it leaves investors indifferent between selling back their shares and holding on to them.

THE MORE GENERAL SOLUTION, GIVEN A BUYBACK PRICE

- Start with the buyback price and compute the **number of shares outstanding after the buyback:**
  - Increase in Debt = Debt at optimal – Current Debt
  - # Shares after buyback = # Shares before – $\frac{\text{Increase in Debt}}{\text{Share Price}}$
- Then **compute the equity value after the recapitalization**, starting with the enterprise value at the optimal, adding back cash and subtracting out the debt at the optimal:
  - Equity value after buyback = Optimal Enterprise value + Cash – Debt
- Divide the equity value after the buyback by the **post-buyback number of shares.**
  - Value per share after buyback = Equity value after buyback/ Number of shares after buyback

LET'S TRY A PRICE: WHAT IF CAN BUY SHARES BACK AT THE OLD PRICE ($67.71)?

- Start with the buyback price and compute the number of shares outstanding after the buyback
  - Debt issued = $ 55,136 - $15,961 = $39,175 million
  - # Shares after buyback = 1800 - $39,175/$67.71 = 1221.43 m
- Then compute the equity value after the recapitalization, starting with the enterprise value at the optimal, adding back cash and subtracting out the debt at the optimal:
  - Optimal Enterprise Value = $153,531
  - Equity value after buyback = $153,531 + $3,931– $55,136 = $102,326
- Divide the equity value after the buyback by the post-buyback number of shares.
  - Value per share after buyback = $102,326/1221.43 = $83.78

BACK TO THE RATIONAL PRICE ($78.61): HERE IS THE PROOF

- Start with the buyback price and compute the number of shares outstanding after the buyback
  - # Shares after buyback = 1800 - $39,175/$78.61 = 1301.65 m
- Then compute the equity value after the recapitalization, starting with the enterprise value at the optimal, adding back cash and subtracting out the debt at the optimal:
  - Optimal Enterprise Value = $153,531
  - Equity value after buyback = $153,531 + $3,931– $55,136 = $102,326
- Divide the equity value after the buyback by the post-buyback number of shares.
  - Value per share after buyback = $102,326/1301.65 = $78.61

2. What if something goes wrong? The Downside Risk

- Sensitivity to Assumptions

  A. "What if" analysis

  - The optimal debt ratio is a function of our inputs on operating income, tax rates and macro variables. We could focus on one or two key variables – operating income is an obvious choice – and look at history for guidance on volatility in that number and **ask what if questions**.

  B. "Economic Scenario" Approach

  - We can develop **possible scenarios**, based upon macro variables, and examine the optimal debt ratio under each one. For instance, we could look at the optimal debt ratio for a cyclical firm under a boom economy, a regular economy and an economy in recession.

- Constraint on Bond Ratings/ Book Debt Ratios
  - Alternatively, we can put **constraints on the bond ratings** to reduce exposure to downside risk. Thus, we could require the firm to have a minimum rating, at the optimal debt ratio or to have a book debt ratio that is less than a "specified" value.

DISNEY'S OPERATING INCOME: HISTORY

Year	*EBIT*	*% Change in EBIT*	*Year*	*EBIT*	*% Change in EBIT*
1987	$756		2001	$2,832	12.16%
1988	$848	12.17%	2002	$2,384	-15.82%
1989	$1,177	38.80%	2003	$2,713	13.80%
1990	$1,368	16.23%	2004	$4,048	49.21%
1991	$1,124	-17.84%	2005	$4,107	1.46%
1992	$1,287	14.50%	2006	$5,355	30.39%
1993	$1,560	21.21%	2007	$6,829	27.53%
1994	$1,804	15.64%	2008	$7,404	8.42%
1995	$2,262	25.39%	2009	$5,697	-23.06%
1996	$3,024	33.69%	2010	$6,726	18.06%
1997	$3,945	30.46%	2011	$7,781	15.69%
1998	$3,843	-2.59%	2012	$8,863	13.91%
1999	$3,580	-6.84%	2013	$9,450	6.62%
2000	$2,525	-29.47%			

Recession	*Δ Operating Income*
2009	Drop of 23.06%
2002	Drop of 15.82%
1991	Drop of 22.00%
1981-82	Increased by 12%
Worst Year	Drop of 29.47%

Standard deviation in % change in EBIT = 19.17%

DISNEY: SAFETY BUFFERS?

EBIT drops by	EBIT	Optimal Debt ratio
0%	$10,032	40%
10%	$9,029	40%
20%	$8,025	40%
30%	$7,022	40%
40%	$6,019	30%
50%	$5,016	30%
60%	$4,013	20%